Cambridge (CIE) A Level Maths: Pure 1

Your notes

Modelling with Trigonometric Functions

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Modelling with trigonometric functions

How do I model with trigonometric functions?

- Various real-life situations can be modelled using trigonometric functions
- You need to be able to interpret the equations used in the model
- If you need to identify maximum or minimum values of a formula, remember the bounds of the **sin** and **cos** functions:
  - $-1 \leq \sin x \leq 1$
  - $-1 \leq \cos x \leq 1$

e.g. WHAT ARE THE MINIMUM AND MAXIMUM VALUES OF $y = 15 - 23\sin 3x$

WHEN $\sin 3x = 1$, $y = 15 - 23(1) = -8$

WHEN $\sin 3x = -1$, $y = 15 - 23(-1) = 38$

SO $y_{max} = 38$ AND $y_{min} = -8$

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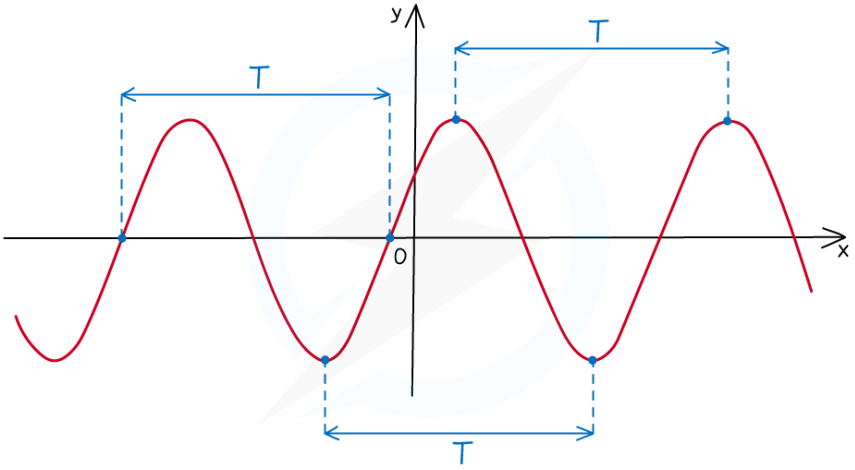
- You may also need to discuss the **period** of an equation
  - The period is often indicated by ***T***
  - For a periodic function in ***x*** like **sin** or **cos**, the **period** is how much ***x*** has to change by for the function to go through one complete cycle

Your notes



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- For functions of the form **cos (*qx*+ *r*)** or **sin (*qx* + *r*)** the period ***T*** is:

$$T = \frac{360}{q} \text{ (for angles in degrees)}$$

$$T = \frac{2\pi}{q} \text{ (for angles in radians)}$$

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e.g. WHAT IS THE PERIOD OF $y = \cos(\frac{7}{8}x - 1.5)$ WHERE THE ANGLE IS MEASURED IN RADIANS?

THE PERIOD IS

$$T = \frac{2\pi}{\frac{7}{8}} = \frac{16\pi}{7}$$

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Examiner Tips and Tricks

- The variable in these questions is often ***t*** for time.
- Read the question carefully to make sure you know what you are being asked to solve.

Worked Example

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An amusement park builds a large circular wheel as a tourist attraction. The height of a passenger on the wheel is modelled by the equation

$$H = 50 - 48\cos\left(\frac{\pi}{15}t + 0.3\right)$$

where H is the height of the passenger above the ground in metres, t is the number of minutes after boarding, and angles are given in radians.

a) What is the maximum height that the passenger reaches above the ground?

b) How many minutes after boarding the ride will the passenger reach that height for the first time?

c) How long does it take the wheel to make a complete revolution?

a) THE MAXIMUM VALUE OF H OCCURS WHEN THE COSINE IS EQUAL TO −1

$H_{max} = 50 - 48(-1) = 98\text{m}$

b) $\cos\left(\frac{\pi}{15}t + 0.3\right) = -1$

$\frac{\pi}{15}t + 0.3 = \cos^{-1}(-1) = \pi$

REMEMBER TO USE RADIANS!

$\frac{\pi}{15}t = \pi - 0.3$

$t = \frac{15}{\pi}(\pi - 0.3) = 13.6$ MINUTES (1 d.p.)

c) THE WHEEL COMPLETES ONE REVOLUTION EACH TIME

$\frac{\pi}{15}t$ GOES THROUGH 2π

$\frac{2\pi}{\left(\frac{\pi}{15}\right)} = 30$ MINUTES

PERIOD OF $\cos(qx+r)$ IS $T = \frac{2\pi}{q}$

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